Name – Soham Debnath

Reg no – 23BAI0084

Speech and Language Processing Lab

BCSE419P

Prof: Dr. Divya Meena Sundaram

Experiment – 2

Aim:

➔ **Identify and explain two problem statements in any domain under Speech and Language Processing systems.**

1. **Problem statement: With widespread use of writing tools that use AI in some form or context, it is becoming increasingly harder to distinguish the differences between AI generated text and genuine human written content. Publishers face difficulty authenticating whether or not a certain article is AI-written or not. Since AI uses text generation and not copy paste from the training source, traditional plagiarism checks often fail.**

   Objective: Build a NLP system that classifies a document as either AI-generated or human written and additionally assign scores based on confidence.

   Existing literature:

   DetectGPT: Avoids training separate classifier and only uses log probabilities computed by model of interest. More discriminative than zero-shot model detection. Computationally expensive and requires two LLM's to function.

   Llm-generated text detection (2025.cl-1.8) – Consolidates research breakthroughs of the year and examines existing datasets. Gap – Out of

Distribution generalization (OOD) – Models struggle with detecting unseen LLMs.

On the possibilities of AI-Generated Text Detection – Argues why and how consistency can be achieved in the task of detecting LLM-generated text and lays groundwork for future research. Tests various generators like Llama-2-13B, GPT-2, GPT-3.5-Turbo etc.

Research Gap: The main obstacle faced by modern state-of-the-art detectors is that real world articles are much messier than the clean benchmark datasets they are trained on – often incorporates mixed (human written + AI generated) text, spelling mistakes, multiple authors, short text, code-mixed languages.

Proposed methodology- To build model for realistic articles, we propose a mixed dataset – a labelled article with clearly marked AI-generated and human written sentences. With techniques such as hard example mining and cross-generator curriculum learning, we slowly skew the model towards more diverse AI-generated content and how to detect them better.

Research Scope – Improve the model further to extract PARTS in an article written by AI, not just classify the article as a whole. Evaluate different pre-trained models on accuracy, recall, F1 score etc.

2. **Problem Statement: Mental Health Detection from Reddit Posts- People often express stress, anxiety, depression and such other states of mind on social media first before seeking professional help. Among the various platforms, Reddit actually sees a larger number of such posts due to its anonymous nature, rather than mainstream platforms like Twitter and Instagram. Automatically identifying such posts could assist in early intervention.**

Objective – Develop an automated bot that identifies and alerts the respective moderators if a user in their thread has displayed signs of these mental illnesses.

Existing literature:

Contextualized Classification of Reddit Posts to DSM-5 for Web-based Intervention – Uses multi-class classifier to catalogue posts to the best matching DSM – 5 (Diagnostic and Statistical Manual of Mental Disorders - 5th Edition). Reduces the false-alarm-rate from 30% to 2.5% over a comparable heuristic baseline, and our mapping results have been verified by domain experts achieving a kappa score of 0.84. However, a heavy reliance on self-reported labels (such as r/depression, r/anxiety etc.) severely downplays its training versatility.

Detection of Mental Health from Reddit via Deep Contextualized Representations – Uses large scale dataset of Reddit posts from users with eight disorders and control user group. Improved over traditional TF-IDF embeddings and Word2Vec methods, but only targets one post at a time – no temporal analysis. Behaves like black boxes.

Research Gap – Limited categories on diseases, English only data training sets, no early risk prediction are the three main problems with the above-mentioned models. Ethical and privacy concerns also arise which are very hard to get rid of.

Proposed methodology – A temporal transformer based model to analyze behaviour in posting frequency and the respective post contents to build a better understanding of the user's risk state.

- ➔ **Preprocess data from collected datasets -> Remove URL's -> Remove usernames -> Emoji normalization -> Sentence segmentation.**
- ➔ **Use models to produce contextual embeddings, such as BERT based models.**
- ➔ **Temporal modelling – Assess risk factor at each post, with later posts getting higher weight in risk if posted more frequently or earlier posts indicate growing unease / symptoms, using BiLSTM.**
- ➔ **Generate explanations using SHAP or Gradients to better categorize the symptoms observed.**
- ➔ **Evaluate accuracy, precision, recall, F1 score etc.**

Future Research scope – cross-platform generalization, multilingual detection are the two most promising aspects of the future research scope in this particular technology.